

DWIJ VYAS

Electronics & Communication Engineering - Embedded Systems, Robotics & Applied AI

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SUMMARY

I'm an Electronics & Communication Engineering undergraduate at BVM, working across embedded systems, digital electronics and applied AI. My work spans hardware design, firmware and the software that ties them together. I'm looking for an internship where I can do this kind of work on a real team.

EDUCATION

Birla Vishvakarma Mahavidyalaya

'24 - now

B.Tech, Electronics & Communication - Vallabh Vidyanagar - CPI 9.05 / 10

Bright Day School

'23 - '24

Higher Secondary, Science - Vadodara - 87.8%

TECHNICAL SKILLS

Embedded: C, C++, ESP32 / ESP8266, ATmega328P, Arduino, Arduino IDE, Sensor fusion, Wireless communication

Robotics: Pixhawk 2.4.8, Raspberry Pi, Jetson Nano, Mission Planner, QGroundControl, Autonomous navigation, Robotics

Digital logic: Verilog, BASYS3 FPGA, Xilinx Vivado, Digital electronics, CPU architecture, Microprocessor/Microcontroller Architecture

Software & AI: Python, MATLAB, C++, Linux / Ubuntu, Autonomous navigation, Sensor fusion

Fabrication: Fusion 360, EasyEDA, PCB design, Bambu Studio, 3D modelling

PROJECTS

TerraAvis - Hybrid drone-tank rescue robot

Robotics - Embedded AI - Pixhawk

An AI-enabled hybrid rescue robot with a dual-mode mobility system combining aerial navigation and tracked ground movement. Integrated thermal imaging, FPV surveillance, real-time object detection and autonomous control for search-and-rescue and disaster-response scenarios.

Technologies: Pixhawk, Raspberry Pi, Python, OpenCV, Thermal imaging, PCB design, Fusion 360, Linux, Autodesk designing, GUI designing, Autonomus Mapping, Mission Planner, ANSYS

Link: <https://drive.google.com/drive/folders/1DYj1aq4hzN4QaE1onNEsXEsNY5stDXUG?usp=sharing>

4-Bit Microprocessor - Built from discrete logic on breadboards

Digital electronics - CPU design

A fully functional 4-bit microprocessor built entirely from discrete 74-series logic ICs on breadboards. Complete datapath including ALU, registers, instruction register, program counter and shared bus, with a hardwired control unit and a custom instruction set supporting arithmetic, logical, load, store and conditional jump operations.

Technologies: 74-series TTL, 555 timer, Breadboard prototyping, Digital Logic Sim

Link: <https://drive.google.com/drive/folders/1dfLMLjH02TEv6RGfnhIRi95xKUdKb2f0?usp=sharing>

Friendly AI Chatbot - Embedded voice assistant

Embedded hardware - AI - PCB design

A standalone embedded AI device built around an ESP32 with real-time voice input, amplified audio output and an OLED display for feedback. Includes a custom-designed PCB and 3D-printed enclosure, with Python driving the conversational logic.

Technologies: ESP32, OLED SSD1306, PAM8403, EasyEDA, Python, Arduino IDE, fusion 360

Link: https://drive.google.com/drive/folders/1A3rikUEOEEWcZX6W_JjXLqyuPbLvr2Cd?usp=sharing

Dual-Axis CNC Plotter - Motion control from G-code to paper

Embedded systems - Mechanical design

A 2-axis CNC plotter using Arduino, stepper motors and a custom mechanical frame to translate digital designs into precise physical drawings. Implemented G-code interpretation and motion-control firmware for accurate pen positioning, combining embedded systems with mechanical design.

Technologies: Arduino, GRBL firmware, Inkscape, Stepper motors, C/C++, Fusion 360

DOSEGUARD - Nuclear Irradiation Decision Engine

AI Systems - Nuclear Science - Interactive Data Design : Based on IAEA

A standards-grounded decision engine for gamma irradiation, built on 30+ IAEA research papers and cross-referenced against each destination country's own regulatory limits (e.g. FDA 21 CFR 179.26). Generates a downloadable, accurate prescription report with tailored recommendations for your exact product and facility. Built for HackAtom Summer University 2026.

Link: <https://6a57ef8a93ac1568f52c2965--unique-torte-e3e753.netlify.app/>

RECOGNITION & EXPERIENCE

Urfu Summer University

Experience - 2026

Summer University Participant - Ural Federal University (URFU), Yekaterinburg, Russia
Gained international-level exposure through URFU's Summer University program and led the HackAtom team in developing DoseGuard, an innovative smart safety solution. Strengthened leadership, teamwork, problem-solving, prototyping, and technical presentation skills in an international academic environment.

Embedded Systems Intern - Modsonic Instruments Mfg. Co. (P) Ltd, Vallabh Vidyanagar

Experience - 2026

Built embedded firmware (C, AVR128DA48) for Modsonic's ultrasonic thickness gauges - upgraded the display/keypad interface to SPI TFT and capacitive touch, and developed an IC LCD driver for the next-gen hardware.

Robofest 5.0 Gujarat - Finalist

Recognition - 2025

TerraAvis, a hybrid drone-tank rescue robot, reached the state finals.